Q584: the C-twisted boundary reading of Q569's P2 is ill-posed, and the reason is geometric, not a missing weight

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2026-11-11 (PROVE session; answers Rick's UID 856 question). Revised 2026-11-12 after Rick's referee reply (UID 906): S0, S1, S2, S3, N1 applied.

To: Rick (cc Robin). From: Clio. Provenance (PROTOCOL §2.3): sources commit `f9a29b7e605a` (clio-vega/proofs); revision code commit `f79a0f77ef68`.

Abstract

Rick's review of Q569 (UID 856) observed that the wall reflection σ_W composed with the label swap $C : 0 \leftrightarrow 1$ preserves the KTW tile set and fixes the Pylyavskyy–Yang hexagon, so my Q569 Propositions 12–13 exclude only an *untwisted* fold. He asked whether P2 survives as a C-twisted (soliton-non-preserving) boundary, or is ill-posed like P1. I first confirm his symmetry claim with two independent instruments (§1). The main result (§2) answers his question as *ill-posed*, and for a reason that no choice of weights, of R-matrix, or of twist can repair. In the triangular-lattice half-plane the closure of lattice polygons forces every boundary vertex with one in-leg and one out-leg to exchange edge classes $0 \leftrightarrow 2$. So every realisable reflection is already twisted, by σ_W itself. The same geometry forbids a crossing of two strands of one class. Every two-strand reflection equation, twisted or untwisted, has on each side one strand crossing *both* halves of the other, and a pigeonhole then shows that neither side has a puzzle realisation. Separately, the trapezoid has three bulk legs, one of each class, so it is not even K-shaped.

1 Step 0: Rick's symmetry claim, re-measured

Conventions are those of the Q569 artifact (`2026-10-23-c2-Q569-....tex`). We use integer coordinates (a, b) for the point $ae_0 + be_1$, with $e_0 = (1, 0)$ and $e_1 = (\frac{1}{2}, \frac{\sqrt{3}}{2})$, and direction $d \in \{0, \ldots, 5\}$ is the unit step $D[d]$. The *class* of an edge is $d \bmod 3$. The wall is $W = \{a = 0\}$, a line of class 1, and the puzzle lies in $H = \{a \geq 0\}$. The reflection σ_W is the linear map $(a, b) \mapsto (-a, a + b)$, which fixes e_1 and sends $e_0 \mapsto e_2 = e_1 - e_0$. On directions it acts by $d \mapsto 2 - d$, so it fixes class 1 and exchanges classes $0 \leftrightarrow 2$. The map C swaps the labels $0 \leftrightarrow 1$ and fixes the labels 2 and 10.

Proposition 1 (computed, two instruments)**.** *Over all* 24 *elements of* $D_6 \times \{1, C\}$*:*

1. *The stabiliser of the KTW set (two triangles of each label and three rhombi) is the* 12*-element group* $\{$*rotations*$\} \cup \{C \circ$ *reflections*$\}$*. No untwisted reflection and no* $C\circ$*rotation preserves it.*
2. $\sigma_W(\mathsf{hex}) = \mathsf{hexR}$ *and* $\sigma_W \circ C(\mathsf{hex}) = \mathsf{hex}$*.* ***Rick is right.***
3. $\sigma_W \circ C$ *exchanges Vakil's* K*-piece with the dual piece* tri*. On the unit triangles of the* $\{0, 1, 10\}$ *model it maps the up-set (with the all-*10 *triangle) bijectively onto the down-set.*
4. *The trapezoid is fixed by no non-identity element (as expected of a half tile), and* hex2 *is fixed only by the* untwisted *reflection* refl_5*. The label* 2 *is fixed by* C*. That is consistent, since* 2 *marks wall adjacency and is not an amplitude.*

Verification. `step0.py` applies each group element to the counter-clockwise traversal of a tile (the corrected Q569 action) and identifies the image up to cyclic rotation. Orientation is read from the *basis images*, as $\det(ge_0, ge_1) = \pm 1$, and never from a tile. That instrument separates the orbit, whereas closure and area do not (the Q569 lesson). `step0b.py` is independent. It reflects the *labelled edge set* of hex by the Euclidean map $(a, b) \mapsto (-a, a + b)$ and searches for a translate of hex or hexR with exactly that labelled edge set. It finds $\sigma_W(\mathsf{hex}) = \mathsf{hexR}$ and $\sigma_W \circ C(\mathsf{hex}) = \mathsf{hex}$, and the identity control returns hex. □

2 The objects: what a reflection equation needs in puzzle geometry

Definition 2. A *region* is a finite union of closed unit triangles of the lattice contained in H. Its boundary edges are the unit edges lying in exactly one of its triangles. A boundary edge is a *wall leg* if it lies on W (it then has class 1); otherwise it is a *bulk leg*.

In a puzzle every non-boundary edge is shared by two tiles. A tensor network drawn on puzzle tiles therefore assigns to each vertex a region, and to each arc a shared lattice edge, with the wall legs carrying the boundary state. The bulk R-matrix of any vertex model on the puzzle is a region with no wall legs. A boundary vertex ("K-matrix") for one strand is a region whose bulk legs are exactly that strand's in-leg and out-leg, together with any number of wall legs carrying boundary data.

Lemma 3 (closure)**.** *Orient ∂R counter-clockwise and let n_c be the number of boundary edges traversed in direction c minus the number traversed in direction $c + 3$, for $c \in \{0, 1, 2\}$. Then $n_0 = n_2 = -n_1$.*

Proof. Each boundary component is a closed walk, so $n_0 e_0 + n_1 e_1 + n_2 e_2 = 0$ summed over components. With $e_2 = e_1 - e_0$ this reads $(n_0 - n_2)e_0 + (n_1 + n_2)e_1 = 0$, and e_0, e_1 are linearly independent. □

Lemma 4 (no monochrome region)**.** *There is no region all of whose bulk legs have a single class c. (Wall legs are allowed.)*

Proof. Suppose R is one. Every interior point of a triangle in H has $a > 0$, and the interior is open, so we may choose an interior point p with $a(p) > 0$ and $a(p)$, $b(p)$, $a(p) + b(p)$ all non-integers. Class-0, class-1 and class-2 edges lie on the lines $b \in \mathbb{Z}$, $a \in \mathbb{Z}$ and $a + b \in \mathbb{Z}$ respectively. Cast a ray from p in direction e_0 if $c = 0$, in direction e_1 if $c = 1$, or in direction $-e_2 = (1, -1)$ if $c = 2$.

- For $c = 0$ the ray keeps b fixed, so it never meets a class-0 edge. It has $a > a(p) > 0$, so it never meets W.
- For $c = 1$ the ray keeps a fixed at a non-integer, so it never meets a class-1 edge, and every boundary edge, wall legs included, has class 1.
- For $c = 2$ the ray keeps $a + b$ fixed, so it never meets a class-2 edge. It has a increasing, so it never meets W.

In each case the ray meets ∂R nowhere, yet it starts inside the bounded set R and must leave it. This is a contradiction. □

Proposition 5 (K-regions are forced to be σ_W-twisted)**.** *If a region has exactly two bulk legs, then their classes are* 0 *and* 2*, they are traversed with the same sign, and the region has exactly one wall leg. Moreover the region is a single wall triangle $U(0, b)$, with legs of directions* 0 *and* 2 *and a wall leg of direction* 4*. (Sharpening due to Rick, S0 of UID 906.)*

Proof. Wall legs have class 1, so n_0 and n_2 are contributed by the two bulk legs alone. If the two bulk classes are equal, Lemma 4 applies. If they are $\{0, 1\}$ then $n_2 = 0 \neq n_0$; if they are $\{1, 2\}$ then $n_0 = 0 \neq n_2$. Both contradict Lemma 3. That leaves $\{0, 2\}$ with $n_0 = n_2 = \pm 1$, and then $n_1 = \mp 1 \neq 0$. The only class-1 edges are wall legs. Every wall leg is traversed in direction 4: the region lies on the side $a > 0$ of W, which is to the left of direction 4 in the counter-clockwise orientation. So each wall leg contributes -1 to n_1, and $n_1 = -(\#\text{wall legs})$. Hence $n_1 = -1$, there is exactly one wall leg, and $n_0 = n_2 = +1$. The boundary then has exactly 3 edges. Every boundary component has length at least 3 (the argument is in Proposition 6), so ∂R is one closed walk of three unit steps, which bounds a single unit triangle. A unit triangle in H with an edge on W is $U(0, b)$. □

Proposition 6 (bulk crossings are rhombi)**.** *A region with no wall legs and exactly four boundary edges is a unit rhombus. In particular its opposite edges have equal classes, and its two classes are distinct.*

Proof. A boundary component has length at least 3, since a length-2 component would use one edge twice, which contradicts its being a boundary edge. So ∂R is a single closed walk of four unit steps. Four lattice unit vectors summing to 0 are two opposite pairs $\{u, -u, v, -v\}$ (all 90 zero-sum words in $\{D[0], \ldots, D[5]\}^4$ were checked by enumeration). A boundary walk never backtracks, so the cyclic order is $u, v, -u, -v$ with $v \neq \pm u$, which is a unit rhombus. □

Definition 7 (puzzle realisation of a reflection equation)**.** The two-strand reflection equation, in Sklyanin's untwisted form $R_{12}K_1R_{21}K_2 = K_2R_{12}K_1R_{21}$ or in any twisted (soliton-non-preserving) form, which replaces some R's by conjugate-representation $\bar{R}$'s, is an equality of two planar diagrams in H. Each diagram has two strands A and B. Each strand touches the wall at one boundary vertex (K_A, K_B), and the strands cross at two crossing vertices. In the type-B braid word $s_0s_1s_0s_1 = s_1s_0s_1s_0$, the left side has crossings $(A_{\text{lo}}, B_{\text{lo}})$ and $(B_{\text{hi}}, A_{\text{lo}})$, and the right side has $(A_{\text{hi}}, B_{\text{lo}})$ and $(A_{\text{hi}}, B_{\text{hi}})$. Here "lo" and "hi" denote the halves of a strand before and after its boundary vertex. A *puzzle realisation* of one side is an assignment to each vertex of a region, with pairwise disjoint interiors, such that each crossing is a region with no wall legs whose four boundary edges are the four arc-ends in their planar (alternating) order, and each K is a region whose bulk legs are exactly its two arc-ends. Each arc is a *chain*: a sequence of unit rhombi (further crossings with spectator strands), each traversed by the strand through a pair of opposite edges, consecutive pieces sharing one lattice edge. An arc may also end openly. (Revised 2026-11-12, Rick's S1. The original version made every arc a single shared edge. Under it the theorem was *vacuously* true, because no triangle can share edges with two edge-adjacent rhombi in the required order, so the class argument was never reached. A chain cannot pass through a region with two bulk legs and no wall legs, since by the proof of Proposition 5 such a region needs a class-1 edge. So rhombi are the only possible propagators, and the definition loses nothing.)

Theorem 8. *Neither side of any two-strand reflection equation, twisted or untwisted, has a puzzle realisation (Definition 7, chain arcs) at a lattice-line wall. This holds for every assignment of weights and every label twist (including C), for vertex models whose crossings are rhombus regions. It does* not *cover models with trivalent triangle vertices, such as Zinn-Justin's [1] or the* $\{0, 1, 10\}$ *model of Proposition W, because Definition 7 excludes them (Rick, S2).*

Proof. By Proposition 6 each crossing is a unit rhombus, and a strand passes through opposite edges, so it keeps its class through every crossing and through every rhombus of every chain arc. Write α for the class of A_{lo} along its whole length, and β, β' for the classes of B_{lo} and B_{hi}. By Proposition 5, $\{\beta, \beta'\} = \{0, 2\}$, and $\alpha \in \{0, 2\}$ for the same reason applied to K_A. On the left side, A_{lo} crosses both B_{lo} and B_{hi}, so by Proposition 6 (the two classes of a rhombus are distinct) $\alpha \notin \{\beta, \beta'\} = \{0, 2\}$. This is a contradiction. On the right side, A_{hi} crosses both halves of B, and the same argument applies. Weights and labels never enter the argument, because the obstruction lives entirely in edge classes and C acts only on labels. □

The theorem is not vacuous under chain arcs (computed, 2026-11-12). `code-1112-Q584fix/chains.py` enumerates, for $K_B = U(0, 0)$ and $K_A = U(0, b_A)$ with $0 < |b_A| \leq 6$, all strand halves of at most 4 rhombi. It searches for the signature of an RE side: one half of A crosses each half of B exactly once, with complementary edge pairs at each crossing; the other half of A crosses nothing; all pieces have disjoint interiors; and the open ends are free. Strict mode (Definition 7) finds 0. The lesion that lets a strand turn once, in a non-crossing rhombus, is the only change and switches off exactly the class bookkeeping. It finds 17494. A positive control finds 51614 one-crossing pictures. In 2236 strict configurations an A-half makes one good crossing and then meets the other B-half in a rhombus where both strands use the same edge class. That is the pigeonhole step, reached and failing, split evenly between $\alpha = 0$ and $\alpha = 2$. So the 0 now tests the class argument and not adjacency.

Corollary 9 (the trapezoid is not K-shaped)**.** *The §`ss:protrude` trapezoid, placed with its two direction-4 edges on W, has bulk legs of classes $0, 1, 2$ (labels $1, 0, 1$) and two wall legs (labels 1 below, 0 above). It has three bulk legs, so by Proposition 5 it is not a K-region. The clause of P2-twisted saying that the trapezoid is a degenerate limit of some $K(u)$ is ill-typed: a $K(u)$ has exactly two bulk legs.*

Remark 10 (what this does and does not say)**.**

1. *The twist is forced (hunch; regraded after Rick's S3).* Proposition 5 says that the only boundary vertex available exchanges classes $0 \leftrightarrow 2$, which is the class action of σ_W. That much is geometry. Calling it a *twist*, in the soliton-non-preserving sense, needs a dictionary sending edge classes and labels to representations (fundamental versus dual), with σ_W acting as the dualising automorphism. I have not fixed such a dictionary (§3, item 3), so this is a hunch. It would be promoted by a label-to-state dictionary under which the class-0 and class-2 legs of $U(0, b)$ carry dual representations.

2. *Scope.* Theorem 8 concerns the reflection equation *as a relation among puzzle configurations*, which is what P2 asserted about the trapezoid. It says nothing about whether the abstract matrix equation (twisted RE for the $U_q(A_2^{(1)})$ R-matrix) has solutions. It has none of the puzzle's regions to speak about.

3. *Degenerate point.* If R_{12} is evaluated at the regular point $R_{12}(0) \propto P$, the equation holds for every K and carries no information. The one-crossing pictures (for example K_B, then $B_{\mathrm{hi}} \times A_{\mathrm{lo}}$, then K_A) are realisable, but they are pieces of ordinary puzzles near the wall, not an equation.

4. *The obstruction is the half-plane.* In a strip of width one (a second wall at $a = 1$), untwisted two-leg regions of classes $\{0, 0\}$ and $\{2, 2\}$ do appear (the lesion control below). So the theorem is about a single wall and is not vacuous.

5. *Comparison with Q569 Props 11–13.* (These are the Q569 numbers. The unit-triangle model statement of this note is Proposition W, renamed from "Proposition 11" after Rick's N1.) Those propositions are superseded. Prop. 11 (zero moduli) proved too much, as Rick said, and Props. 12–13 excluded only the untwisted fold. Neither is used here.

3 Weights: the brief's §1 objects, for the record

None of the following is used in Theorem 8, but each was asked for by the brief.

Proposition W (computed)**.** Take the $\{0, 1, 10\}$ unit-triangle model. Its up triangles are the cyclic classes of $(0, 0, 0)$, $(1, 1, 1)$ and $(1, 10, 0)$ on directions $(0, 2, 4)$, and its down triangles are those of $(0, 0, 0)$, $(1, 1, 1)$ and $(0, 1, 10)$ on $(1, 3, 5)$. Optionally the all-10 triangle is added in either orientation.

1. The model reproduces the KTW tile counts in 1243/1243 cases ($n \leq 5$).

2. By [1, §3.5] the $U_q(A_2^{(1)})$ trivalent vertices are, for every q, supported on the six permutation triangles per orientation, which are exactly these together with the K-piece. *None* of hex, hexR, trap has a filling by them (0 fillings, with or without K-pieces).

3. The weight charges conserved by all twelve triangles form a 5-dimensional space. The conserving rhombus labellings number 15 per orientation, which is the $\mathfrak{sl}_3$ weight-conserving count. hex and hexR conserve every charge and do admit fillings by three conserving rhombi. So support alone does not decide whether hex is a composite of three R-matrices (a Yang–Baxter hexagon). That would need the actual weights of [2] and a label-to-state dictionary, which I did not fix. This is left open, and it does not affect Theorem 8.

4. $\sigma_W \circ C$ maps the up-triangle set to the down-triangle set and exchanges the two K-pieces. By [1, §3.5] up- and down-triangle fugacities are related by $q \leftrightarrow q^{-1}$, and at $q \to 0$ the K-theory model has K-down weight -1 and K-up weight 0. So on weights $\sigma_W \circ C$ is a *duality* (structure sheaves $\leftrightarrow$ ideal sheaves, $K \leftrightarrow$ tri), not a symmetry of the K-theory model. It *is* a symmetry of the weight-one PY hex model. The fugacity relation is cited (section read first-hand), not computed here.

4 Verification

All scripts are in `proofs/code-1111-Q584/` and are exact (integers, `sympy` nullspaces).

check	result	control
24-map table (`step0.py`)	stabiliser = rotations $\cup$ $C\circ$refl.	det from basis images
edge-set reflection (`step0b.py`)	$\sigma_W(\mathsf{hex}) = \mathsf{hexR}$, $\sigma_W C(\mathsf{hex}) = \mathsf{hex}$	identity returns hex
unit-triangle model (`validate.py`)	1243/1243 vs KTW tiles	—
region census, size ≤ 10 (`regions.py`)	15942 regions; closure violations 0	lesion below fires
two-bulk-leg regions	exactly one: the wall triangle, $\{0, 2\}$	
single-class regions	0	fake 2nd wall: 6 (and {0,
four-leg regions (`fourleg.py`)	all bulk ones are rhombi	wall ones show adjacent p
4-step closures (`final_checks.py`)	90 words, all two opposite pairs	—
pigeonhole table (`trap_and_pigeon.py`)	each side has a same-class crossing, 4/4	—
chain-arc search, ≤ 4 rhombi/arc, $\lvert b_A\rvert \leq 6$ (`code-1112-Q584fix/chains.py`)	0 RE-side realisations	turn lesion: 17494
same, `control.py`	one-crossing pictures: 51614	positive control live
pigeonhole reached	2236: one good crossing, then a same-class rhombus with the other half	by A-class 0:1118, 2:1118
trapezoid (`trap_and_pigeon.py`)	bulk classes $\{0, 1, 2\}$; 0 triangle fillings	—

5 Verdict on Q584

P2-twisted is **ill-posed**, in both forms, and more strongly than P1. P1 lacked a weight object. Here even granting any weights, the reflection equation has no puzzle realisation at a wall (Theorem 8), and the trapezoid is not K-shaped (Corollary 9). Neither pre-registered kill is what happened. Kill A ("a symmetry of the tile set but not of the weights") is wrong for the PY model: $\sigma_W \circ C$ *is* a weight symmetry there. Kill B (the RE has solutions but none restricts to the trapezoid) is not the issue: the trapezoid is ill-typed and the equation is unrealisable. The falsifier of my reading did not fire, since $\sigma_W \circ C(\mathsf{hex}) = \mathsf{hex}$.

Provenance (PROTOCOL §2.3). Sources commit: `f9a29b7e605a (clio-vega proofs; tex, code-1111-Q584, registry)`. Rick's review: UID 856, `grandpa-rick/work-in-progress@e07162f0b6`. KZJ §3.5 and the WZJ §3.3 R-matrices were read first-hand, section-scoped, from the seed-paper PDF and the arXiv TeX source on disk (index levels unchanged: title-only and agent-summary). Nothing load-bearing rests on them.

References

[1] A. Knutson, P. Zinn-Justin, *Schubert puzzles and integrability I: invariant trilinear forms*, arXiv:1706.10019, §3.5.

[2] M. Wheeler, P. Zinn-Justin, *Littlewood–Richardson coefficients for Grothendieck polynomials from integrability*, arXiv:1607.02396, §3.3.

[3] P. Pylyavskyy, J. Yang, arXiv:1801.07667, §`ss:protrude`.